

Entity Relationship Diagram
and
Relational Schema

Relational Schema:

- Airline (airline_id: char(2), name: varchar(100), primary key (airline_id))
- Aircraft (aircraft_id: integer, model: varchar(100), manufacturer: varchar(100), total_seats: integer, econ_capacity: integer, bus_capacity: integer, first_capacity: integer, airline_id: char(2), primary key (aircraft_id), foreign key (airline_id) references Airline (airline_id))
- Airport (airport_code: char(3), name: varchar(100), city: varchar(100), country: varchar(100), primary key (airport_code))
- Airline_Airport (airline_id: char(2), airport_code: char(3), primary key (airline_id, airport_code), foreign key (airline_id) references Airline (airline_id), foreign key (airport_code) references Airport (airport_code))
- Flight (flight_number: varchar(10), airline_id: char(2), aircraft_id: integer, departure_airport: char(3), arrival_airport: char(3), departure_time: time, arrival_time: time, arrival_date: date, flight_date: date, flight_type: varchar(15), primary key (flight_number, airline_id), foreign key (airline_id) references Airline (airline_id), foreign key (aircraft_id) references Aircraft (aircraft_id), foreign key (departure_airport) references Airport (airport_code), foreign key (arrival_airport) references Airport (airport_code))
- Flight_Days (flight_number: varchar(10), airline_id: char(2), day_of_week: varchar(15), primary key (flight_number, airline_id, day_of_week), foreign key (flight_number, airline_id) references Flight (flight_number, airline_id))
- Customer (customer_id: integer, first_name: varchar(50), last_name: varchar(50), email: varchar(100), phone: varchar(20), address: varchar(255), dob: date, username: varchar(50), password: varchar(255), account_creation_date: date, primary key (customer_id))
- Employee (employee_id: integer, first_name: varchar(50), last_name: varchar(50), email: varchar(100), phone: varchar(20), username: varchar(50), password: varchar(255), role: varchar(25), primary key (employee_id))

→ Ticket(ticket_number: integer, customer_id: integer, total_fare: float, quantity: integer, purchase_datetime: datetime, ~~ticket_type: varchar(10)~~, is_flexible: boolean, status: varchar(15), primary key (ticket_number), foreign key (customer_id) references Customer(customer_id), booked_by: integer, foreign key (booked_by) references Employee(employee_id))

→ Ticket-Segment(ticket_number: integer, sequence_number: integer, flight_number: varchar(10), airline_id: char(2), flight_date: date, seat_number: varchar(10), class: varchar(10), special_meal: boolean, from_airport: char(3), to_airport: char(3), primary key (ticket_number, sequence_number), foreign key (ticket_number) references Ticket(ticket_number), foreign key (flight_number, airline_id) references Flight(flight_number, airline_id), foreign key (from_airport) references Airport(airport_code), foreign key (to_airport) references Airport(airport_code))

→ Waiting-List(waitlist_id: integer, customer_id: integer, flight_number: varchar(10), airline_id: varchar(5), flight_date: date, class: varchar(20), waitlist_position: integer, added_date: timestamp, primary key (waitlist_id), foreign key (customer_id) references Customer(customer_id), foreign key (flight_number, airline_id) references Flight(flight_number, airline_id))

→ QA_Entry(qa_id: integer, customer_id: integer, question_text: text, answer_text: text, created_at: timestamp, foreign key (customer_id) references Customer(customer_id), primary key (qa_id))

Group 36

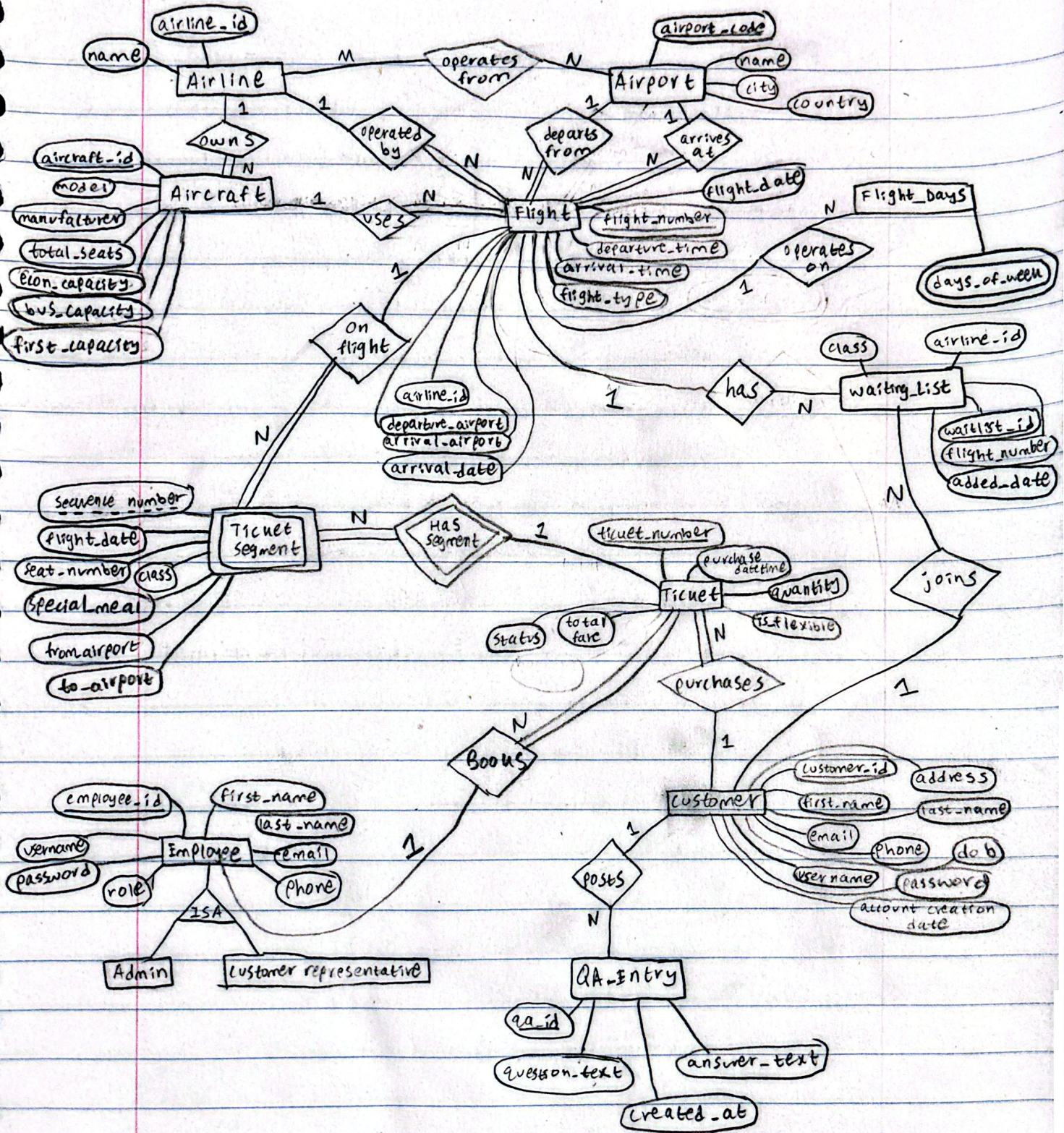
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UPDATED

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Entity Relationship Diagram

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